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Knowledge Extraction Using Aspect-Based Sentiment Analysis and Classification Method

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Abstrak— **In this modern era, the use of technology in various fields of life is growing, especially in the financial sector such as digital payments. Digital payments are transactions made via the internet with various forms or electronic payment applications. This research utilizes one of the e-wallet applications, namely the DANA Application. As one of the digital payment applications, DANA's service quality is an important factor in competing with various e-wallet applications or other competitive digital payment applications, sentiment analysis of massive amounts of incoming review data is needed to understand user perceptions and satisfaction levels in order to help companies in decision making. This research analyzes 69632 user review data from May 1 to June 28, 2024, then classifies reviews into positive, negative, and neutral reviews based on aspects to find out the dominant aspects in the review data, as for the aspects determined in this study are mobile applications, interfaces, service performance, and security. This research compares various classification method algorithms to determine the performance of the model, the comparison results show that the Random Forest algorithm has the best performance of other algorithms studied based on evaluation metrics with accuracy 0.82, precision 0.82, recall 0.82, f1_score 0.82, AUC-ROC 0.92, and Cross Validation Mean Accuracy 0.822. This research is expected to produce useful and computerized knowledge extraction and provide perceptual understanding, user satisfaction, and insight into the most effective algorithms which can then help companies in making decisions in the future.**

Keyword— **Knowledge Extraction, ABSA, Classification, DANA.**

I. INTRODUCTION

At this time the use of technology in various fields is growing in Indonesia, especially in the financial sector such as digital payments [1]. Digital payment is a transaction made via the internet with various forms or electronic payment applications [2]. This research utilizes one of the well-known electronic payment applications or digital wallets in Indonesian society [3], namely the DANA application, the application allows users to perform digital payment services.

DANA is an e-wallet application that provides financial services [1]. As one of the digital payment applications, DANA's service quality is an important factor in competing with various other competitive e-wallet or digital payment applications [1]. Service quality analysis can take advantage of data mining by conducting sentiment analysis of reviews that arrive massively on various online platforms, especially on Play Store, sentiment analysis is needed to obtain valuable information related to user perceptions and satisfaction levels with DANA services in a relatively fast time, especially if the data used includes large data [4].

The technique of sentiment analysis is used to analyze public opinion and perception of services or products [5]. Sentiment analysis is very important in analyzing data because it can help related parties in decision making [6], the sentiment analysis process results in the classification of opinion text or reviews related to services or products into positive, negative, and neutral sentiments [4].

Among the most widely used classification algorithms is Naïve Bayes because it is straightforward and uncomplicated [7]. The Naïve Bayes algorithm is employed for classification using the Bayes theorem which is a fundamental principle in probability theory [8]. Benefits Naïve Bayes can estimate classification-related parameters with minimal training data [9].

An method called Support Vector Machine (SVM) can handle data in various dimensions and has the ability to find the optimal decision boundary for the division of different sentiment classes by searching for the optimal or best hyperlane among an unlimited number of functions by maximizing the distance between classes [7] using the principle of structural risk minimization (SRM).

Logistic Regression is a statistical method that calculates a feature's outcome and determines the likelihood that it is part of a binary class [10]. Logistic Regression is known as

a log-linear classifier, which is used to model the possible outcomes of model efforts [11].

Random Forest is an algorithm used for classification and regression that consists of many trained decision trees, the classification results are based on the estimated average of the decision trees [11]. This algorithm uses an overall score by comparing the facts that occur with estimates from the decision sub-tree data set [12]

In previous research, there are several related studies regarding sentiment analysis of the DANA application including by [1], the study processed user review data from OVO and DANA services. From the testing of various algorithms, the highest accuracy was obtained by the K-NN, which scored 91%, followed by Random Forest (87%), Naïve Bayes (87%) and Decision Tree (86%).

In different research [3], it was done by processing review data from e-wallet applications OVO and DANA. Testing was done with several algorithms and using ten cross-validations. The study revealed that OVO is the application with the most users and performance models with a 91% accuracy of the SVM model and an AUC value of 0.986.

A study by [13] using various algorithms to conduct sentimental analysis, a study conducted to examine user data from several digital wallets in Indonesia, one of which is DANA. This study evaluates the performance and accuracy of various algoritms. This research showed that the SVM algorithm has an accuracy of 92%, the Naïve Bayes Multinominal model has a rate of accuration of 88%, and the Random Forest model achieves an accurateness of about 85%.

The next research by [14], sentiment analysis was carried out by utilizing Naïve Bayes Classification. The research processed 11,885 review data to analyze the sentiment of DANA service quality based on several dimensions. The research shows that 69% of reviews are positive, and less than 42% of reviews are negative, and the dominant issue lies in the dimension of customer compensation.

Furthermore, research by [15] conducted sentiment analysis using deep learning methods. Research processes review data from several e-wallet applications including DANA, sentiment analysis is carried out based on several categories, namely transactions, access, services, accounts, and performance. The results showed that the review data tended to be negative with 87% model accuracy and 83% model validation.

Based on the problems and previous research, there has been no research that focuses on producing knowledge extraction for the DANA application, and the use of data in previous studies averaged below 10,000 data. Knowledge extraction is important in various applications, including natural language processing, social media analysis, document processing, and other applications that require handling large amounts of unstructured information to be converted into useful knowledge and can be processed computerized [16], [17], [18]. So that sentiment analysis in this study will focus on producing knowledge extraction on massive incoming review data with a large amount of 69632 data.

Sentiment analysis is carried out using the classification method and utilizing the ABSA method which will classify data into negative, positive, and neutral reviews based on several aspects to determine the dominant aspects in the review, as for the predetermined aspects, namely aspects of mobile applications, interfaces, service performance, and security. This research also compares several algorithms used to evaluate how well the model [12] performs in sentiment analysis related to DANA reviews, so as to provide information related to understanding user perceptions and satisfaction as well as insight into the most effective algorithms which can then help companies make decisions in the future [5].

II. METHODS

In Figure 1 there are research stages or research steps that have been developed from previous research [11]. The research process begins with data collection, data cleaning, pre-processing data, aspect-based sentiment analysis (ABSA), data labeling, train-test split, model testing (model evaluation, model validation, error analysis), model evaluation based on evaluation matrix, model validation using Stratified K-Fold Cross Validation, model error analysis using confusion matrix, knowledge extraction, visualization of results from several algorithms, and conclusion.

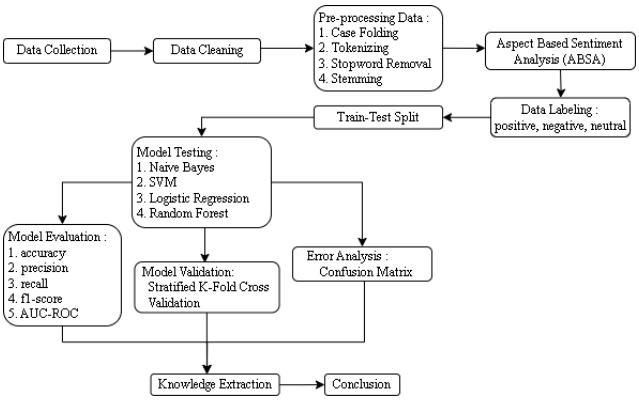


Figure 1. Research stages [12]

A. Data Collection

In this step, the web scraping technique is carried out, namely the process of retrieving review data in the PlayStore application with the help of google play scraper [11], this process retrieves 69632 data with a time span from May 1 to June 28 in the year 2024 related to DANA application reviews, the retrieval is done using the phyton language by utilizing the google play scraper API in the use of google colaboratory. On June 29 2024, the data retrieval process was carried out, the data that was successfully retrieved will be transferred in Comma Separated Value or CSV format.

TABLE I
DATA SCRAPING RESULT

Username	Review	Rating	Date
Google User	ok, very good and satisfied, easy and simple	5	6/27/2024 04:10
Google User	when I topped up my balance of 30,000 last afternoon, when I wanted to make a transaction, my balance was sucked up until it ran out, please, what's the problem?	1	6/23/2024 04:28
Google User	dana error	2	6/24/2024 11:24
Google User	Hello admin, after updating, why now when the transfer is finished you can't SHARE the proof of transfer so you have to take a screenshot now	3	6/08/2024 02:07
Google User	The application is ok but it's a shame that every time you top up there are lots of deductions	4	6/22/2024 06:41

Based on Table I, there are 5 examples of data among the 69632 review data that was successfully obtained, these reviews were obtained using google collaboratory with the phyton programming language, typing errors in the reviews have been corrected using slang. In the table above there are user names, reviews, ratings, and dates.

B. Data Cleaning

The data cleaning process, starting with selecting the columns to be analyzed, cleaning from duplicate data, handling missing values and outliers, removing special symbols, removing numbers, removing leading and trailing spaces and excessive spaces, and removing rows with empty content.

TABLE II
DATA CLEANING RESULT

Username	Review	Rating	Date
Google User	ok very good and satisfied easy and simple	5	6/27/2024 04:10
Google User	when I topped up my balance of last afternoon when I wanted to make a transaction my balance was sucked up until it ran out please whats the problem	1	6/23/2024 04:28
Google User	dana error	2	6/24/2024 11:24
Google User	Hello admin after updating why now when the transfer is finished you cant SHARE the proof of transfer so you have to take a screenshot now	3	6/08/2024 02:07
Google User	The application is ok but its a shame that every time you top up there are lots of deductions	4	6/22/2024 06:41

The dataset is cleaned and prepared using data cleaning before further analysis so that it can provide important information only [2]. After the cleaning process is carried out, the scraping data which initially amounted to 69632 data was reduced to 68686 data. In Table II there are 5 examples of data that have gone through the cleaning process and are ready for further analysis.

C. Pre-processing Data

The data pre-processing stage is the next stage after cleaning [19]. Pre-processing data is used so that the cleaned data is easy to process and understand so that it facilitates the classification process. The data pre-processing stage involves a number of procedures, including folding cases, tokenization, removing stopword, and stemming [20].

1) Case Folding : During the process, every capital character in the review data is converted to a lowercase letter [21], the process is carried out in order to avoid differences in uppercase and lowercase letters that can cause errors in analysis.

TABLE III
CASE FOLDING RESULT

Review
ok very good and satisfied easy and simple
when i topped up my balance of last afternoon when i wanted to make a transaction my balance was sucked up until it ran out please whats the problem
dana error
hello admin after updating why now when the transfer is finished you cant share the proof of transfer so you have to take a screenshot now
the application is ok but its a shame that every time you top up there are lots of deductions

In Table III there are 5 examples of data used as samples, the data is data that has gone through the case folding stage, so it has passed the stage of converting uppercase letters into lowercase letters.

2) Tokenizing : Tokenizing is a process to break down review data or sentences into several words or tokens [22], the tokenizing process helps prepare data for further analysis.

TABLE IV
TOKENIZING RESULT

Review
[ok, very, good, and, satisfied, easy, and, simple]
[when, i, topped, up, my, balance, of, last, afternoon, when, i, wanted, to, make, a, transaction, my, balance, was, sucked, up, until, it, ran, out, please, whats, the, problem]
[dana, error]
[hello, admin, after, updating, why, now, when, the, transfer, is, finished, you, cant, share, the, proof, of, transfer, so, you, have, to, take, a, screenshot, now]
[the, application, is, ok, but, its, a, shame, that, every, time, you, top, up, there, are, lots, of, deductions]

In Table IV there are 5 examples of data used as samples, the data is data that has gone through the tokenizing stage, so it has passed the stage where the sentence is divided into several words.

3) Stopword Removal : One technique to reduce or eliminate words is called stopword removal that do not contribute much such as articles and prepositions [23], the stopword removal process helps reduce words that have no influence in text analysis.

TABLE V
STOPWORD REMOVAL RESULT

Review
[ok, good, satisfied, easy, simple]
[topped, balance, last, afternoon, wanted, make, transaction, balance, sucked, ran, please, whats, problem]
[dana, error]
[hello, admin, updating, transfer, finished, cant, share, proof, transfer, take, screenshot]
[the, application, ok, shame, every, time, top, lots, deductions]

In Table V there are 5 examples of data used as samples, the data is data that has gone through the stopword removal stage, so it has passed the stage of removing common words that are not important.

4) Stemming : Stemming is a process used to restore words that have been transformed into words in the shape of their origins in the preceding stage, thus helping the preparation of texts, words, and documents [24].

TABLE VI
STEMMING RESULT

Review
[ok, good, satisfied, easy, simple]
[top, balance, last, afternoon, want, make, transaction, balance, suck, ran, please, what, problem]
[dana, error]
[hello, admin, update, transfer, finish, cant, share, proof, transfer, take, screenshot]
[the, application, ok, shame, every, time, top, lot, deduction]

In Table VI there are 5 examples of data used as samples, the data is data that has gone through the stemming stage, so it has passed the stage of returning the word to its root form.

D. Aspect Based Sentiment Analysis (ABSA)

An analysis technique called Aspect Based Sentiment Analysis (ABSA) looks at text-related factors of a document and assigns each aspect a sentiment level [25]. Models for aspect-based categorization are trained to determine an aspect's sentiment polarity [26]. This research uses several aspects to the sentiment of review data to determine customer satisfaction.

TABLE VII
ASPECT FOR ANALYSIS

No	Aspect	Word in Teks
1	Mobile Apps	transaction', 'transfer', 'top up', 'customer service', 'payment', 'app', 'mobile', 'upgrade', 'update', 'renew', 'premium', 'pay', 'qris', 'paylater', 'withdraw', 'take', 'funds', 'phone', 'download', 'install', 'send', 'atm', 'account', 'cashback', 'purchase', 'verify', 'easy', 'cost', 'admin', 'install', 'heavy', 'user', 'update', 'device', 'android', 'live chat', 'account', 'help', 'install', 'voucher'
2	Interface	'interface', 'interface', 'display', 'ui', 'easy', 'easy', 'history', 'ad', 'complicated', 'pop up', 'hassle', 'confused', 'column', 'feature', 'usage', 'menu', 'doubt'
3	Service Performance	'performance', 'performance', 'service', 'service', 'force close', 'fast', 'good', 'steady', 'long', 'difficult', 'good', 'good', 'okay', 'ok', 'error', 'system', 'exit', 'connection', 'network', 'login', 'pin', 'glitch', 'loading', 'useful', 'reliable', 'helpful', 'star', 'error', 'server', 'bug', 'login', 'expired', 'scan', 'barcode', 'lag', 'constraint', 'slow', 'fail', 'disappointed', 'wrong', 'bad', 'broken', 'difficult', 'cool', 'best', 'crashing', 'complain', 'complaint', 'smooth', 'hot', 'fix', 'response'
4	Security	'security', 'balance', 'money', 'danger', 'lost', 'cash', 'safe', 'fake', 'tighten', 'approval', 'hack', 'breach', 'suck', 'siphon', 'hacked', 'steal', 'broken into', 'strong', 'leak', 'data', 'otp', 'unscrupulous', 'conceded', 'confirm', 'password', 'report', 'hack'

Based on Table VII, there are aspects that researchers will use to determine customer satisfaction, each aspect has several words that will analyze the text in the DANA review.

E. Data Labeling

Data labeling is the stage of giving positive, negative, and neutral sentiment values [27]. In determining the sentiment label for DANA review data based on predetermined aspects, for positive and neutral sentiments, researchers utilize composite sentiment scores from the use of *Sentiment Intensity Analyzer* which is a lexicon-based sentiment analysis tool from the NLTK library. As for neutral reviews, if the composite score does not provide enough information, the function will consider the review rating [28].

TABLE VIII
DATA LABELING USING ABSA

Review	Rating	Aspect	Sentiment Score	Sentiment
ok very good and satisfied easy and simple	5	interface, service performance	{"interface": 0.296, "service performance": 0.296}	Positive
when I topped up my balance of last afternoon when I wanted to make a transaction my balance was sucked up until it ran out please whats the problem	1	mobile apps, security	{"mobile apps": -0.3612, "security": -0.3612}	Negative
dana error	2	mobile apps, service performance	{"mobile apps": -0.4019, "service performance": -0.4019}	Negative
Hello admin after updating why now when the transfer is finished you cant SHARE the proof of transfer so you have to take a screenshot now	3	mobile apps, interface	{"mobile apps": 0.0, "interface": 0.0}	Neutral
The application is ok but its a shame that every time you top up there are lots of deductions	4	mobile apps, service performance	{"mobile apps": 0.4588, "service performance": 0.4588}	Positive

In Table VIII there are 5 examples of data that are used as samples, the data is data that has gone through the data labeling stage with the use of ABSA, labeling is done by utilizing compound scores and ratings on reviews.

F. Split Data

Split data is a stage carried out by dividing data into small and meaningful units to be analyzed in data processing [29]. This study's data divided into portions with a testing to training data ratio of 70:30.

TABLE VIX
TRAIN-TEST SPLIT

Comparison Ratio	Data Training	Data Testing
70 : 30	48080	20606

Based on Table VIX, there are the results of the split data that has been carried out. To complete the separation, the reviews is divided utilizing a 70:30 ratio, into the 48080 data train and the 20606 data test.

G. Model Testing

Several classification techniques are used in model testing, Random Forest, Naive Bayes, Support Vector Machine (SVM), and Logistic Regression are among them. Model evaluation, result validation, and model error analysis were used in this study's model testing.

1) Model Evaluation : In this study, model evaluation is performed by utilizing the use of an evaluation matrix to assess the performance of the model using a number of related classification techniques, including f1_score, AUC-ROC, accuracy, precision, and recall [30].

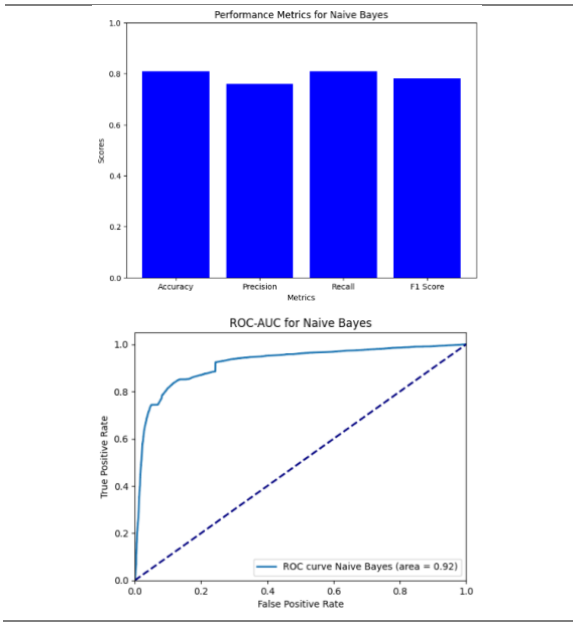


Figure 3. Naïve bayes performance and AUC-ROC

The results of the outcomes of the evaluation of the Naïve Bayes model are displayed using the evaluation matrix in Figure 3. According to the evaluation results, The AUC-ROC of the Naïve Bayes model is 0.92, the accuracy is 0.81, there is a 0.76 precision, a 0.81 recall, and a 0.76 f1 score.

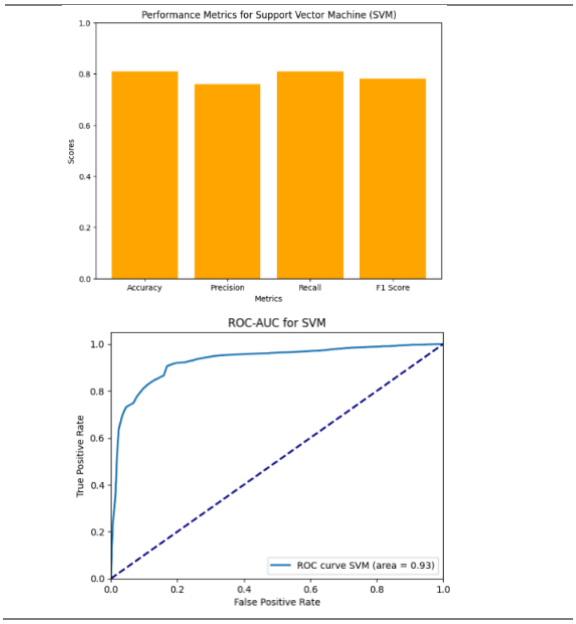


Figure 4. Support vector machine (SVM) performance and AUC-ROC

The Support Vector Machine evaluation (SVM) model's results are visualized using the evaluation matrix in Figure 4. Evaluation findings show that the SVM model has 0.91 AUC-ROC, a recall of 0.81, an accuracy of 0.91, a precision of 0.76, and a f1 score of 0.76.

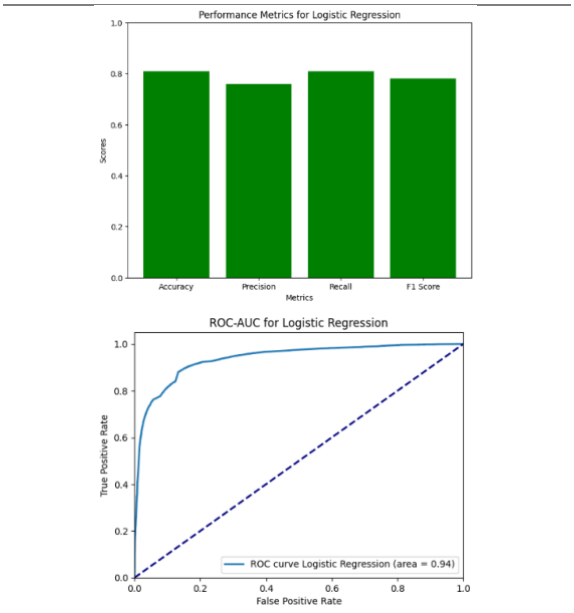


Figure 5. Logistic Regression Performance and AUC-ROC

Figure 5 shows the evaluation results of the Logistic Regression model, which are based on the evaluation matrix. The evaluation findings indicate that the Logistic Regression model possesses the following metrics: accuracy 0.81, there is 0.76 precision, a 0.81 recall, a 0.76 f1_score, and AUC-ROC 0.94.

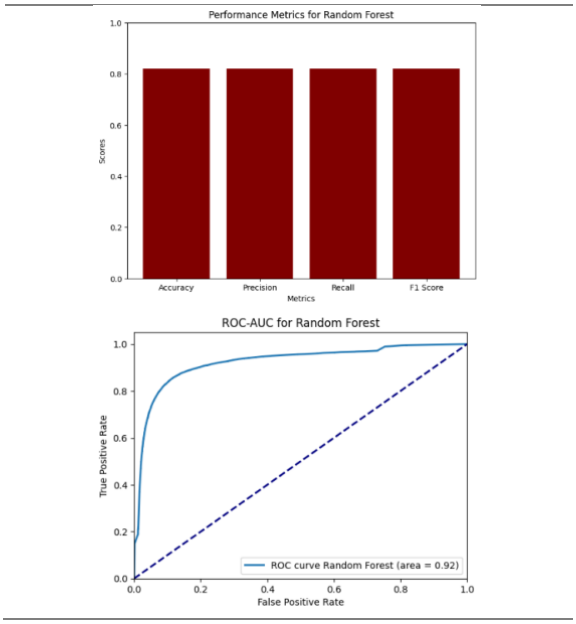


Figure 6. Random Forest Performance and AUC-ROC

In Figure 6 there is a visualization of the Random Forest model's evaluation outcomes based on the evaluation matrix. According to the evaluation results, the Random Forest model has 0.92 AUC-ROC, 0.82 precision, 0.82 recall, 0.82 f1_score, and 0.82 accuracy.

2) Model Validation : This study validates models associated with several classification models that use a form It is also called as K-fold cross-validation in stratified. Stratified Cross Validation using K-folds is an augmentation method that divides data into train and test sets into several subsets or folds with class distribution in all folds, so that the proportion for each class is the same [31]. The following is model validation using Stratified K-fold Cross Validation for each model.

Naive Bayes Classification Report:

	precision	recall	f1-score	support
negative	0.72	0.81	0.76	5208
neutral	0.13	0.02	0.03	1755
positive	0.85	0.91	0.88	13643
accuracy			0.81	20606
macro avg	0.57	0.58	0.56	20606
weighted avg	0.76	0.81	0.78	20606

Accuracy: 0.8075803164126953
Cross-validation scores for Naive Bayes: [0.80084437 0.79617093 0.80956541 0.81982966 0.80
Mean cross-validation score for Naive Bayes: 0.8070058841905325

Figure 7. Naïve bayes model validation

SVM Classification Report:

	precision	recall	f1-score	support
negative	0.76	0.74	0.75	5208
neutral	0.12	0.03	0.04	1755
positive	0.84	0.93	0.88	13643
accuracy			0.81	20606
macro avg	0.57	0.57	0.56	20606
weighted avg	0.76	0.81	0.78	20606

Accuracy: 0.8071920799767058
Cross-validation scores for SVM: [0.79735041 0.79209434 0.81196768 0.81917449 0.8230326
Mean cross-validation score for SVM: 0.8087239229756873

Figure 8. Support vector machine (SVM) model validation

Logistic Regression Classification Report:

	precision	recall	f1-score	support
negative	0.76	0.74	0.75	5208
neutral	0.13	0.03	0.05	1755
positive	0.84	0.93	0.88	13643
accuracy			0.81	20606
macro avg	0.58	0.57	0.56	20606
weighted avg	0.76	0.81	0.78	20606

Accuracy: 0.8078229641851887
Cross-validation scores for Logistic Regression: [0.7982239 0.79369586 0.81415156 0.82303269 0.8
Mean cross-validation score for Logistic Regression: 0.8111944183751256

Figure 9. Logistic regression model validation

Random Forest Classification Report:

	precision	recall	f1-score	support
negative	0.72	0.79	0.75	5208
neutral	0.55	0.48	0.51	1755
positive	0.90	0.88	0.89	13643
accuracy			0.82	20606
macro avg	0.72	0.72	0.72	20606
weighted avg	0.82	0.82	0.82	20606

Accuracy: 0.8224788896437931
Cross-validation scores for Random Forest: [0.81547532 0.80585281 0.82194074 0.83649996 0.8308
Mean cross-validation score for Random Forest: 0.8221814114691844

Figure 10. Random forest model validation

The use of Stratified Cross Validation is generally with k between 5 and 10 [8]. Based on the results of the model validation evaluation above, this study uses K with 5 folds to produce a Cross Validation Mean Accuracy value for each model. The Cross Validation Mean Accuracy value for Naive Bayes is 0.807, whereas the SVM model gets a value of 0.809, for the Logistic Regression model 0.811, and the last model is Random Forest with a value of 0.822.

3) Model Error Analysis : This research analyzes errors related to some classification model performance using Confusion Matrix [19]. A technique for analyzing model errors called the confusion matrix can give a thorough knowledge of the mistakes the model makes and how the values of false, true negative, false negative, and true positive are used to base the error distribution [32].

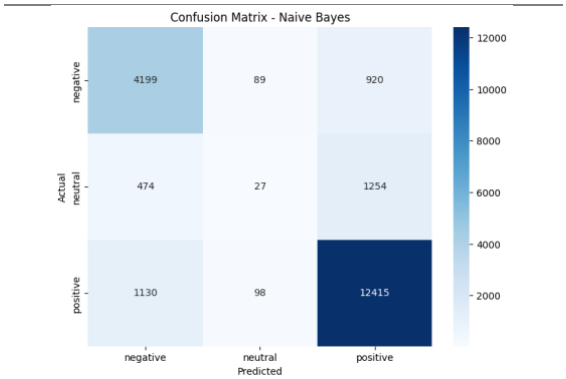


Figure 11. Naïve bayes

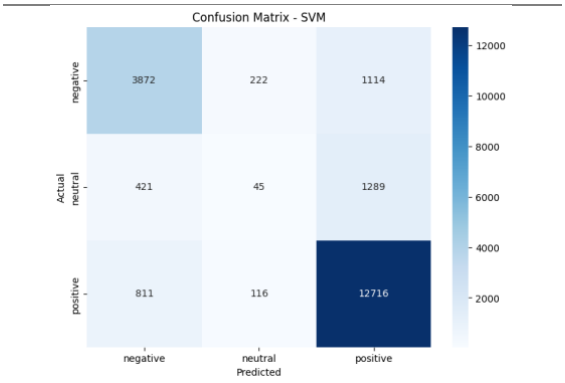


Figure 12. Support vector machine (SVM)

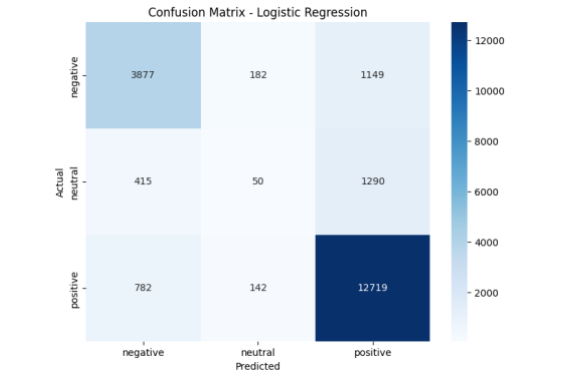


Figure 13. Logistic regression

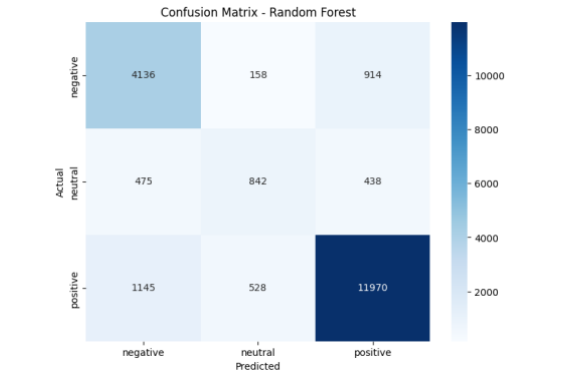


Figure 14. Random forest

In the figure above there are the results of model performance in predicting or analyzing errors from various classification method algorithm models, based on Figure 11 the Naive Bayes model successfully predicts 12415 word values with positive sentiment, 27 word values with neutral sentiment, and 4199 word values with negative sentiment. Then Figure 12 SVM model successfully predicts 12719 word values with positive sentiment, 45 word values with neutral sentiment, and 3877 word values with negative sentiment. In Figure 13 the Logistic Regression model successfully predicts 12719 word values with positive sentiment, 50 word values with neutral sentiment, and 3877 word values with negative sentiment. Furthermore, Figure 14 Random Forest model successfully

predicts 11970 word values with positive sentiment, 842 word values with neutral sentiment, and 4136 word values with negative sentiment.

III. Result And Discussion

A. *Sentiment Analysis*

Sentiment analysis conducted using the classification method by utilizing Aspect Based Sentiment Analysis (ABSA) successfully classifies data into several aspects, namely mobile apps, interface, service performance, and security. The details of the classification process in using ABSA in review text analysis for labeling consider the review rating to get better results. The review text is rated based on the combined score, reviews with a combined score of more than 0.05 are considered a positive sentiment, if the combined score is negative are those less than -0.05, a combined score of 0.00 then the function will look at the rating aspect of the review, If the review receives a score of three, the sentiment is neutral, a score of less than three indicates a negative feeling, and a score of more than three indicates a positive sentiment.

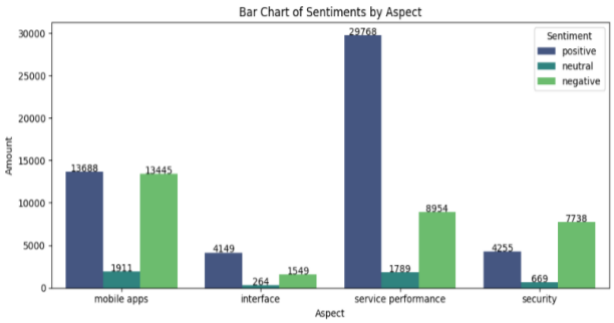


Figure 15. DANA review sentiment by aspect

Based on Figure 15, there is a visualization of sentiment results based on aspects of the sentiment analysis process that has been carried out, in these results positive sentiment, neutral sentiment, and negative sentiment are obtained in each aspect. The mobile apps aspect obtained 13688 positive reviews, 1911 neutral reviews, and 13445 negative reviews. Interface aspects obtained 4149 positive reviews, 264 neutral reviews, and 1549 negative reviews. Service performance aspects get 29768 positive reviews, 1789 neutral reviews, and 8954 negative reviews. The security aspect is 4255 positive reviews, 669 neutral reviews, and 7738 negative reviews. In addition to the sentiment results based on aspects, this study has an overall sentiment distribution with a total of 45475 positive reviews, 5850 neutral reviews, and 17361 negative reviews.

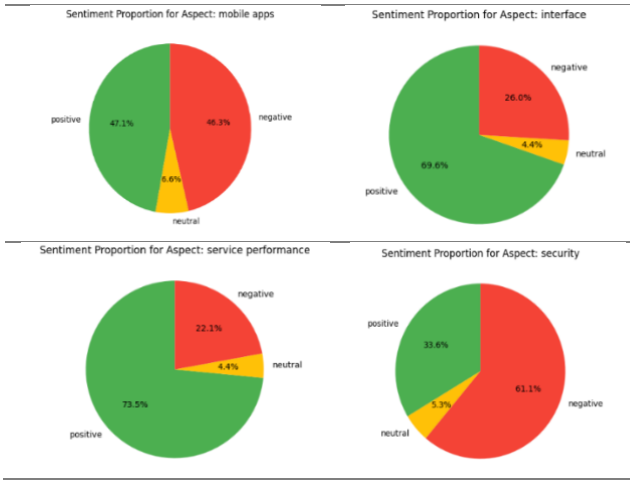


Figure 16. Sentiment proportion

In Figure 16 there is a visualization of the sentiment proportion used to find the dominant aspect based on the percentage of positive, neutral, and negative sentiment results in each aspect. Based on the sentiment proportion, it can be concluded that the dominant issue in the review is the security of the DANA application, this is evidenced by the proportion of sentiment on the security aspect having the highest negative percentage of 61.1%.

B. *Classification Model Testing*

After Sentiment Analysis is carried out, it is followed by testing several algorithms of classification methods. Model testing is carried out based on model evaluation with evaluation matrices such as f1_score, precision, recall, and accuracy, AUC-ROC. In addition to model evaluation, this research uses five folds of stratified cross validation to validate the model, in order for that it will produce the average accuracy of cross validation.

TABLE X
Classification Model Testing Result

Algorithm	Accuracy	Precision	Recall	F1_Score	AUC-ROC	Cross Validation Mean Accuracy
Naive Bayes	0.80	0.76	0.81	0.78	0.92	0.807
Support Vector Machine (SVM)	0.81	0.76	0.81	0.78	0.93	0.809
Logistic Regression	0.81	0.76	0.81	0.78	0.94	0.811
Random Forest	0.82	0.82	0.82	0.82	0.92	0.822

Based on Table X, there are results of model testing of various algorithms from the classification method, the value in these results will be used to find the best algorithm for the process of sentiment analysis related to DANA reviews. The information in the table will be visualized in the form of performance images such as Figure 17 to make it easier to read the data and find the best algorithm.

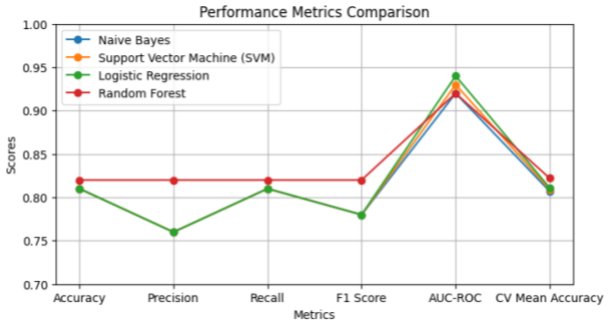


Figure 17. Visualization of model testing results

Figure 17 is a visualization of the test results of various classification method algorithms. As can be observed from the graph, the figure compares several algorithm models depending on the assessment metrics generated earlier, compared to other algorithm models, Random Forest has better and more consistent performance as evidenced by higher accuracy, precision, recall, f1_score, and cross validation mean accuracy values than other algorithms, Logistic Regression excels at AUC-ROC but overall performance is still below Random Forest.

C. Knowledge Extraction

Knowledge Extraction is a process carried out to uncover and extract information and knowledge [33]. In the ABSA results which show that among the several aspects analyzed, the security aspect is the aspect that produces the highest level of negative sentiment, namely the security aspect, this shows that the satisfaction of DANA application users is still very low regarding security compared to other aspects. Furthermore, the outcomes of model testing that demonstrate the Random Forest model's superiority and are based on evaluation, validation, and error analysis.



Figure 18. Knowledge extraction from sentiment analysis of DANA app reviews

Knowledge extraction for sentiment analysis has been investigated by [34] and [35]. These studies implement knowledge extraction in the context of textual data processing to analyze sentiment, but with a focus on different techniques, such as research [34] which uses RDF graphs for evaluation of the quality of extraction results, while research [35] emphasizes more on processing

product reviews using NLP. Based on Figure 18, this study adopts a similar technique by performing knowledge extraction for sentiment analysis. However, the analysis results are converted into XML format, which allows storing ABSA results and model testing in a structured manner.

IV. CONCLUSION

In the sentiment analysis research on DANA application reviews from PlayStore was conducted using Aspect-based Sentiment Analysis (ABSA) and model testing using a number of classification methods, such as Support Vector Machine, Random Forest, Logistic Regression, and Naive Bayes, to generate Knowledge Extraction. This research generates sentiment based on aspects of mobile apps, interface, service performance, and security, based on 69632 DANA evaluate the data, the findings indicate that the highest negative percentage of 61.1% is in the security aspect, which is evidence that the security section is a dominant issue in user satisfaction related to DANA application reviews. Furthermore, the test results of model evaluation and model validation of sentiment results, the evaluation results are depending on the AUC-ROC, f1_score, accuracy, precision, recall, while model validation is carried out by utilizing Stratified K-Fold Cross Validation, besides that this study also uses Confusion Matrix to predict or analyze errors in sentiment results. Model testing results demonstrate that the Random Forest algorithm is a very strong and reliable model compared to other algorithms analyzed, and in the confusion matrix results the Random Forest model also has superior detection capabilities. Use of the Random Forest algorithm is possible as the best choice in classification related to DANA reviews and can then be utilized by agencies as a consideration in decision making, especially related to dominant issues in reviews.

The study's findings provide recommendations for additional research to better understand and overcome the limitations of this study, such as exploring other models, or conducting a grid search to identify the most suitable hyperparameter combination for each model, expanding the scope of analysis by analyzing several applications related to digital payments or digital wallets, as well as exploring broader and more recent data.

REFERENCES

[1] I. Tahyudin, A. R. Hananto, S. A. Rahayu, R. M. Anjani, and A. Nurhopipah, "Sentiment Analysis Model Development on E-Money Service Complaints," TEM Journal, vol. 12, no. 4, pp. 2050–2055, Nov. 2023, doi: 10.18421/TEM124-15.

[2] H. Wisnu, M. Afif, and Y. Ruldevyani, "Sentiment analysis on customer satisfaction of digital payment in Indonesia: A comparative study using KNN and Naïve Bayes," J Phys Conf Ser, vol. 1444, no. 1, p. 012034, Jan. 2020, doi: 10.1088/1742-6596/1444/1/012034.

[3] [3] D. A. Kristiyanti, D. A. Putri, E. Indrayuni, A. Nurhadi, and A. H. Umam, "E-Wallet Sentiment Analysis Using Naïve Bayes and Support Vector Machine Algorithm," J Phys Conf Ser, vol. 1641, no. 1, p. 012079, Nov. 2020, doi: 10.1088/1742-6596/1641/1/012079.

[4] A. Alsokkar, M. Otair, H. E. Alfar, A. Y. Nasereddin, K. Aldiabat, and L. Abualigah, "Sentiment analysis for Arabic call center notes using machine learning techniques," Journal of Autonomous Intelligence, vol. 7, no. 3, Jan. 2024, doi: 10.32629/jai.v7i3.940.

[5] Y. Teek Bau, T. Eng Leong, and C. Le Goh, "Sentiment Analysis of E-Wallet Companies: Exploring Customer Ratings and Perceptions," Journal of Logistics, Informatics and Service Science, vol. 10, no. 4, Nov. 2023, doi: 10.33168/JLISS.2023.0413.

[6] L. Z. S. Sudar, J. L. Imbenay, I. Budi, A. Ramadiah, P. K. Putra, and A. B. Santoso, "Textual Analysis for Public Sentiment Toward National Police Using CRISP-DM Framework," Revue d'Intelligence Artificielle, vol. 38, no. 1, pp. 63–72, Feb. 2024, doi: 10.18280/ria.380107.

[7] M. Y. Al Qahar, Y. Ruldeviyani, U. N. Mukharomah, M. A. Fidyawan, and R. Putra, "Factor analysis influencing Mobile JKN user experience using sentiment analysis," IAES International Journal of Artificial Intelligence (IJ-AI), vol. 13, no. 2, p. 1782, Jun. 2024, doi: 10.11591/ijai.v13.i2.pp1782-1793.

[8] A. Sabir, H. A. Ali, and M. A. Aljabery, "ChatGPT Tweets Sentiment Analysis Using Machine Learning and Data Classification," Informatica, vol. 48, no. 7, May 2024, doi: 10.31449/inf.v48i7.5535.

[9] Khyrina Airin Fariza Abu Samah, Nur Shahirah Jailani, Raseeda Hamzah, Raihah Aminuddin, Nor Afirdaus Zainal Abidin, and Lala Septem Riza, "Aspect-Based Classification and Visualization of Twitter Sentiment Analysis Towards Online Food Delivery Services in Malaysia," Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 37, no. 1, pp. 139–150, Jan. 2024, doi: 10.37934/araset.37.1.139150.

[10] M. Tabany and M. Gueffal, "Sentiment Analysis and Fake Amazon Reviews Classification Using SVM Supervised Machine Learning Model," Journal of Advances in Information Technology, vol. 15, no. 1, pp. 49–58, 2024, doi: 10.12720/jait.15.1.49-58.

[11] T. V. Sai Krishna, T. S. Rama Krishna, S. Kalime, C. V. Murali Krishna, S. Neelima, and R. R. PBV, "A novel ensemble approach for Twitter sentiment classification with ML and LSTM algorithms for real-time tweets analysis," Indonesian Journal of Electrical Engineering and Computer Science, vol. 34, no. 3, p. 1904, Jun. 2024, doi: 10.11591/ijeecs.v34.i3.pp1904-1914.

[12] N. Z. Al Habesyah, R. Herteno, F. Indriani, I. Budiman, and D. Kartini, "Sentiment Analysis of TikTok Shop Closure in Indonesia on Twitter Using Supervised Machine Learning," Journal of Electronics, Electromedical Engineering, and Medical Informatics, vol. 6, no. 2, pp. 148–156, Apr. 2024, doi: 10.35882/jeeemi.v6i2.381.

[13] B. Harnadi and A. D. Widiantoro, "Evaluating the Performance and Accuracy of Supervised Learning Models on Sentiment Analysis of E-Wallet," in 2023 7th International Conference on Information Technology (InCIT), IEEE, Nov. 2023, pp. 175–180. doi: 10.1109/InCIT60207.2023.10413111.

[14] S. Zahra and A. Alamsyah, "Digital Wallet Service Quality Analysis using Multiclass Classification and Sentiment Analysis," in 2022 5th International Conference on Information and Communications Technology (ICOIACT), IEEE, Aug. 2022, pp. 256–261. doi: 10.1109/ICOIACT55506.2022.9971936.

[15] A. A. Fattahila, F. Irfan Amorokhman, K. M. Auditama, K. Ahmad Wijaya, and A. Romadhony, "Indonesian Digital Wallet Sentiment Analysis Using CNN And LSTM Method," in 2021 International Conference on Artificial Intelligence and Big Data Analytics, IEEE, Oct. 2021, pp. 1–6. doi: 10.1109/ICAIBDA53487.2021.9689712.

[16] G. K. Palshikar et al., "RINX: A system for information and knowledge extraction from resumes," Data Knowl Eng, vol. 147, p. 102202, Sep. 2023, doi: 10.1016/j.datak.2023.102202.

[17] T. S. Sonnenschein, G. A. de Wit, N. R. den Braver, R. C. H. Vermeulen, and S. Scheider, "Validating and constructing behavioral models for simulation and projection using automated knowledge extraction," Inf Sci (N Y), vol. 662, p. 120232, Mar. 2024, doi: 10.1016/j.ins.2024.120232.

[18] V. Muppavarapu, G. Ramesh, A. Gyrard, and M. Noura, "Knowledge extraction using semantic similarity of concepts from Web of Things knowledge bases," Data Knowl Eng, vol. 135, p. 101923, Sep. 2021, doi: 10.1016/j.datak.2021.101923.

[19] N. Aisyah Binti Zakaria Adli, M. Ahmad, N. Abdul Ghani, S. Devi Ravana, and A. Anir Norman, "An Ensemble Classification of Mental Health in Malaysia Related to the Covid-19 Pandemic Using Social Media Sentiment Analysis," Ksii Transactions on Internet and Information Systems, Vol. 18, No. 2, Feb. 2024, Doi: 10.3837/Tiis.2024.02.006.

[20] P. K. Jain, R. Pamula, and G. Srivastava, "A Systematic Literature Review on Machine Learning Applications for Consumer Sentiment Analysis Using Online Reviews," Comput Sci Rev, Vol. 41, P. 100413, Aug. 2021, Doi: 10.1016/J.Cosrev.2021.100413.

[21] M. Taufik Hidayat and T. Mauritsius, "Perception of The Indonesian Society on The Performance of The Ministry of Investment/Bkpm Based on Sentiment Analysis," International Journal of Engineering Trends and Technology, Vol. 72, No. 2, Pp. 242–253, Feb. 2024, Doi: 10.14445/22315381/Ijett-V72i2p125.

[22] T. T. Mengistie and D. Kumar, "Deep Learning Based Sentiment Analysis on Covid-19 Public Reviews," In 2021 International Conference on Artificial Intelligence in Information and Communication (ICAIIC), IEEE, 2021, Pp. 444–449.

[23] N. A. Alabdulkarim, M. A. Haq, and J. Gyani, "Exploring Sentiment Analysis on Social Media Texts," Engineering, Technology & Applied Science Research, Vol. 14, No. 3, Pp. 14442–14450, Jun. 2024, Doi: 10.48084/Etasr.7238.

[24] E. Qaqish, A. Aranki, and W. Etaiwi, "Sentiment Analysis and Emotion Detection of Post-Covid Educational Tweets: Jordan Case," Soc Netw Anal Min, Vol. 13, No. 1, P. 39, Mar. 2023, Doi: 10.1007/S13278-023-01041-8.

[25] Y. C. Chang, C. H. Ku, and D. D. Le Nguyen, "Predicting Aspect-Based Sentiment Using Deep Learning and Information Visualization: The Impact of Covid-19 on The Airline Industry," Information and Management, Vol. 59, No. 2, Mar. 2022, Doi: 10.1016/J.Im.2021.103587.

[26] N. Li, X. Yang, I. A. Wong, R. Law, J. Y. Xu, and B. Zhang, "Automating Tourism Online Reviews: A Neural Network Based Aspect-Oriented Sentiment Classification," Journal of Hospitality and Tourism Technology, Vol. 14, No. 1, Pp. 1–20, Jan. 2023, Doi: 10.1108/Jhtt-03-2021-0099.

[27] T. Hamonangan Saragih and D. Kartini, "Analisis Sentimen Brand Ambassador BTS Terhadap Tokopedia Menggunakan Klasifikasi Bayesian Network Dengan Ekstraksi Fitur Tf-IDF," JIP (Jurnal Informatika Polinema), 2023.

[28] O. Alqaryouti, N. Siyam, A. Abdel Monem, and K. Shaalan, "Aspect-Based Sentiment Analysis Using Smart Government Review Data," Applied Computing And Informatics, Vol. 20, No. 1/2, Pp. 142–161, Jan. 2024, Doi: 10.1016/J.Aci.2019.11.003.

[29] R. K. Das, M. Islam, M. M. Hasan, S. Razia, M. Hassan, and S. A. Khushbu, "Sentiment Analysis in Multilingual Context: Comparative Analysis of Machine Learning And Hybrid Deep Learning Models," Heliyon, Vol. 9, No. 9, P. E20281, Sep. 2023, Doi: 10.1016/J.Heliyon.2023.E20281.

[30] H. Swapnarekha, J. Nayak, H. S. Behera, P. B. Dash, and D. Pelusi, "An Optimistic Firefly Algorithm-Based Deep Learning Approach for Sentiment Analysis of Covid-19 Tweets," Mathematical Biosciences and Engineering, Vol. 20, No. 2, Pp. 2382–2407, 2022, Doi: 10.3934/Mbe.2023112.

[31] E. Pujo Ariesanto Akhmad, K. Adi, and A. Puji Widodo, "Machine Learning Approach to Customer Sentiment Analysis in Twitter Airline Reviews," E3s Web of Conferences, Vol. 448, P. 02044, Nov. 2023, Doi: 10.1051/E3sconf/202344802044.

[32] M. Maree, M. Eleyat, and E. Mesqali, "Optimizing Machine Learning-Based Sentiment Analysis Accuracy in Bilingual Sentences Via Preprocessing Techniques," The International Arab Journal of Information Technology, Vol. 21, No. 2, 2024, Doi: 10.34028/Iajit/21/2/8.

[33] G. Ciaburro and G. Iannace, "Machine Learning-Based Algorithms to Knowledge Extraction from Time Series Data: A Review," Data (Basel), Vol. 6, No. 6, P. 55, May 2021, Doi: 10.3390/Data6060055.

[34] A. Gangemi, D. R. Recupero, M. Mongiovì, A. G. Nuzzolese, and V. Presutti, "Identifying Motifs for Evaluating Open Knowledge Extraction on The Web," Knowl Based Syst, Vol. 108, Pp. 33–41, Sep. 2016, Doi: 10.1016/J.Knosys.2016.05.023.

[35] Y. Jiao and Q.-X. Qu, "A Proposal for Kansei Knowledge Extraction Method Based on Natural Language Processing Technology and Online Product Reviews," Comput Ind, Vol. 108, Pp. 1–11, Jun. 2019, Doi: 10.1016/J.Compind.2019.02.011.